

The master z -integral, from scratch

$$T^{mm'} = \int d^d z \frac{(y-z)^m (y'-z)^{m'}}{(y-z)^2 (y'-z)^2} \text{ — two independent derivations, every step}$$

$r \equiv y' - y, \hat{r} \equiv r/|r|, r^2 \equiv |r|^2, d = 2 \text{ (cutoff) or } d = 2 - 2\epsilon \text{ (dimensional regularisation)}$

The answer, twice.

$$\text{cutoff } |u| < R : \quad T^{mm'} = \frac{\pi}{2} \left(\log \frac{R^2}{r^2} + 1 \right) \delta^{mm'} - \pi \hat{r}^m \hat{r}^{m'}$$

$$d = 2 - 2\epsilon : \quad T^{mm'} = \pi \mathcal{N}(\epsilon) (\pi \mu^2 r^2)^{-\epsilon} \left[\frac{\delta^{mm'}}{2\epsilon} - \hat{r}^m \hat{r}^{m'} \right], \quad \mathcal{N}(\epsilon) = \Gamma(1 + \epsilon) \frac{\Gamma(1 - \epsilon)^2}{\Gamma(2 - 2\epsilon)}$$

The $\hat{r}^m \hat{r}^{m'}$ coefficient is $-\pi$ in both — it is finite, so it cannot depend on the regulator, and that it comes out identical by two completely different routes is the check that both derivations are right.

0. Where it comes from, and what it physically is

In your row, after the Wilson-line reduction, the point z (the unobserved gluon's transverse position) appears in **exactly two places**: the two $\mathcal{C}$ -vertex kernels

$$\frac{(y-z)^m}{(y-z)^2} \quad \text{and} \quad \frac{(y'-z)^{m'}}{(y'-z)^2}.$$

$U(z)$ has cancelled, the phase does not contain z , and $\rho(x), \rho(x')$ do not either. So the z integral factorises off completely and is the object above. Nothing else in the row has to be touched to do it.

What it is physically. $\frac{z^m}{z^2}$ is the Weizsäcker-Williams kernel, and up to constants it is the two-dimensional Fourier transform of $\frac{\ell^m}{\ell^2}$ — the amplitude to emit a gluon of transverse momentum ℓ . So

$$T^{mm'} \sim \int d^2 \ell \frac{\ell^m \ell^{m'}}{(\ell^2)^2} e^{i\ell \cdot r},$$

which is logarithmically divergent as $\ell \rightarrow 0$: the integrand goes as $\frac{1}{\ell^2}$ against $\ell d\ell$. **Small ℓ is large $|z|$, and small relative transverse momentum is what "collinear" means.** That is why the pole you are about to find is a collinear pole and gets absorbed into a fragmentation function — not a short-distance (UV) pole.

1. Setup

Shift the integration variable. Put

$$u \equiv y - z \implies z = y - u, \quad d^d z = d^d u, \quad y' - z = (y' - y) + u = u + r.$$

$$T^{mm'} = \int d^d u \frac{u^m (u+r)^{m'}}{u^2 (u+r)^2}$$

Two facts to keep in hand:

- **It is symmetric**, $T^{mm'} = T^{m'm}$. Send $u \rightarrow -u - r$: the measure and both denominators are invariant, and the numerator becomes $(u+r)^m u^{m'}$. So swapping the indices gives back the same integral.
- **It can only be built from $\delta^{mm'}$ and $r^m r^{m'}$** , because after integrating over u the only vector left is r . So $T^{mm'} = A \delta^{mm'} + B \hat{r}^m \hat{r}^{m'}$ with A, B scalars. Everything below is just working out A and B .

2. Power counting — find the divergence *before* you integrate

Always do this first; it tells you which regulator you need and what the answer must look like.

region	integrand	measure	verdict
$u \rightarrow 0$	$\frac{u^m r^{m'}}{u^2 r^2} \sim \frac{1}{ u }$	$ u d u $	$\int d u $ — finite
$u \rightarrow -r$	same by symmetry	$\rho d\rho$	finite
$ u \rightarrow \infty$	$\frac{u^m u^{m'}}{u^4} \sim \frac{1}{ u ^2}$	$ u d u $	$\int \frac{d u }{ u }$ — LOG DIVERGENT

So there is exactly one divergence and it is at **large** $|u|$. Two ways to regulate it:

1. **Cut off** at $|u| < R$. Then the answer will contain $\log R^2$.
2. **Work in** $d = 2 - 2\epsilon$ **with** $\epsilon > 0$. At large $|u|$ the radial measure is $|u|^{d-1} d|u|$, so the integral goes as $\int^\infty |u|^{d-3} d|u| = \int^\infty |u|^{-1-2\epsilon} d|u|$, which **converges for** $\epsilon > 0$.

Sign convention — read this twice. Here $\epsilon > 0$ regulates a **large-distance** divergence. The $\frac{1}{\epsilon}$ you will get is an **infrared / collinear** pole, even though the symbol looks like the familiar UV pole of four-dimensional loops. Do not let the notation fool you into thinking you have found a UV divergence: the power counting in §2 already proved there is none at short distance.

3. Method 1 — $d = 2$ with a cutoff, by complex coordinates

In two dimensions this integral is elementary if you use complex coordinates, and you learn what each piece of the answer is doing.

3.1 Complex notation

For a two-vector $V^m = (V^1, V^2)$ write

$$V \equiv V^1 + iV^2, \quad \bar{V} \equiv V^1 - iV^2, \quad V^2 = V\bar{V}, \quad a \cdot b = \frac{1}{2}(a\bar{b} + \bar{a}b).$$

The one identity that makes everything work: the WW kernel splits into a holomorphic and an antiholomorphic pole,

$$\frac{a \cdot u}{u^2} = \frac{\frac{1}{2}(a\bar{u} + \bar{a}u)}{u\bar{u}} = \frac{1}{2} \left(\frac{a}{u} + \frac{\bar{a}}{\bar{u}} \right).$$

3.2 Contract with two arbitrary constant vectors

Rather than chase indices, compute $a_m b_{m'} T^{mm'}$ for arbitrary a, b and read off the tensor at the end:

$$\begin{aligned} a_m b_{m'} T^{mm'} &= \frac{1}{4} \int d^2 u \left(\frac{a}{u} + \frac{\bar{a}}{\bar{u}} \right) \left(\frac{b}{u+r} + \frac{\bar{b}}{\bar{u}+\bar{r}} \right) \\ &= \frac{1}{4} \int d^2 u \left[\underbrace{\frac{ab}{u(u+r)}}_{(i)} + \underbrace{\frac{a\bar{b}}{u(\bar{u}+\bar{r})}}_{(ii)} + \underbrace{\frac{\bar{a}b}{\bar{u}(u+r)}}_{(iii)} + \underbrace{\frac{\bar{a}\bar{b}}{\bar{u}(\bar{u}+\bar{r})}}_{(iv)} \right]. \end{aligned}$$

So everything reduces to two master integrals, one "holomorphic-holomorphic" and one "mixed".

3.3 Master 1: $\int_{|u|<R} \frac{d^2 u}{u-a} = -\pi \bar{a}$ (for $|a| < R$)

Split the disc at $|u| = |a|$ and expand in each region. Write $u = \rho e^{i\phi}$, so $d^2 u = \rho d\rho d\phi$ and $u^n = \rho^n e^{in\phi}$.

Outer region $|a| < |u| < R$. Expand in a/u :

$$\frac{1}{u-a} = \frac{1}{u} \frac{1}{1-a/u} = \sum_{n \geq 0} \frac{a^n}{u^{n+1}}.$$

Every term has angular dependence $e^{-i(n+1)\phi}$ with $n+1 \geq 1$, and $\int_0^{2\pi} e^{-i(n+1)\phi} d\phi = 0$. **The whole outer region gives zero.**

Inner region $|u| < |a|$. Expand in u/a :

$$\frac{1}{u-a} = -\frac{1}{a} \frac{1}{1-u/a} = -\frac{1}{a} \sum_{n \geq 0} \frac{u^n}{a^n}.$$

Now $\int_0^{2\pi} e^{in\phi} d\phi = 0$ except for $n = 0$, which survives and gives the area: $\int_{|u| < |a|} d^2u = \pi|a|^2$. Hence

$$\int_{|u| < R} \frac{d^2u}{u-a} = -\frac{1}{a} \pi|a|^2 = -\pi \frac{a\bar{a}}{a} = -\pi\bar{a}. \quad \checkmark$$

Sanity check at $a = 0$: $\int \frac{d^2u}{u} = \int \rho d\rho \int e^{-i\phi} d\phi = 0 = -\pi\bar{0} \quad \checkmark$. Numerically at $a = 0.7 + 0.3i$, $R = 60$: $-2.2189 + 0.9510i$ against the exact $-2.1991 + 0.9425i \quad \checkmark$.

3.4 Master 2: $\int_{|u| < R} \frac{d^2u}{(\bar{u}-\bar{a})(u-b)} = \pi \log \frac{R^2}{|a-b|^2}$

Two standard two-dimensional facts, with $\partial_{\bar{u}} = \frac{1}{2}(\partial_{u^1} + i\partial_{u^2})$:

$$(F1) \quad \partial_{\bar{u}} \log |u-a|^2 = \frac{1}{\bar{u}-\bar{a}}, \quad (F2) \quad \partial_{\bar{u}} \frac{1}{u-b} = \pi \delta^{(2)}(u-b).$$

(F1) is immediate from $\log |u-a|^2 = \log(u-a) + \log(\bar{u}-\bar{a})$, since $\partial_{\bar{u}}$ annihilates the first. (F2) is the two-dimensional Green's function identity — the same statement as $\nabla^2 \log |u|^2 = 4\pi \delta^{(2)}(u)$.

Use (F1) to write the integrand as a total derivative minus a remainder:

$$\frac{1}{(\bar{u}-\bar{a})(u-b)} = \partial_{\bar{u}} \left[\frac{\log |u-a|^2}{u-b} \right] - \log |u-a|^2 \partial_{\bar{u}} \frac{1}{u-b}.$$

Integrate over the disc. For the first piece use Stokes in the form $\oint_{\partial D} F du = 2i \int_D \partial_{\bar{u}} F d^2u$, i.e. $\int_D \partial_{\bar{u}} F d^2u = \frac{1}{2i} \oint F du$:

$$\frac{1}{2i} \oint_{|u|=R} \frac{\log |u-a|^2}{u-b} du \xrightarrow{R \gg |a|, |b|} \frac{\log R^2}{2i} \oint_{|u|=R} \frac{du}{u-b} = \frac{\log R^2}{2i} (2\pi i) = \pi \log R^2.$$

For the second, (F2) collapses the integral onto $u = b$:

$$\begin{aligned} & - \int d^2u \log |u-a|^2 \pi \delta^{(2)}(u-b) = -\pi \log |b-a|^2. \\ \implies & \int_{|u| < R} \frac{d^2u}{(\bar{u}-\bar{a})(u-b)} = \pi \log \frac{R^2}{|a-b|^2}. \quad \checkmark \end{aligned}$$

Numerically at $R = 60$: 23.770 against the exact 23.793 $\checkmark$.

3.5 Apply the masters to the four terms

term	rewritten	master	value
(i) $\frac{1}{u(u+r)}$	$\frac{1}{r} \left[\frac{1}{u} - \frac{1}{u+r} \right]$	Master 1 twice, with $a = 0$ and $a = -r$	$\frac{1}{r} [0 - \pi\bar{r}] = -\pi \frac{\bar{r}}{r}$
(ii) $\frac{1}{u(u+r)}$	$\frac{1}{(\bar{u}-(-r))(u-0)}$	Master 2, $a = -r$, $b = 0$	$\pi \log \frac{R^2}{r^2}$
(iii) $\frac{1}{\bar{u}(u+r)}$	$\frac{1}{(\bar{u}-0)(u-(-r))}$	Master 2, $a = 0$, $b = -r$	$\pi \log \frac{R^2}{r^2}$
(iv) $\frac{1}{\bar{u}(u+r)}$	complex conjugate of (i)	—	$-\pi \frac{r}{\bar{r}}$

Partial fractions used in (i): $\frac{1}{u(u+r)} = \frac{1}{r} \left[\frac{1}{u} - \frac{1}{u+r} \right]$, and $\int \frac{d^2u}{u+r} = \int \frac{d^2u}{u-(-r)} = -\pi \overline{(-r)} = \pi\bar{r}$.

$$a_m b_{m'} T^{mm'} = \frac{\pi}{4} \left[-ab \frac{\bar{r}}{r} - \bar{a}\bar{b} \frac{r}{\bar{r}} + (a\bar{b} + \bar{a}b) \log \frac{R^2}{r^2} \right].$$

3.6 Back to real notation

Two translations. The easy one is $a\bar{b} + \bar{a}b = 2(a \cdot b)$. For the other, write $\frac{\bar{r}}{r} = \bar{\hat{r}}^2$ and $\frac{r}{\bar{r}} = \hat{r}^2$ (using $\hat{r}\bar{\hat{r}} = 1$), and expand the projection onto $\hat{r}$:

$$(a \cdot \hat{r})(b \cdot \hat{r}) = \frac{1}{4} (a\bar{\hat{r}} + \bar{a}\hat{r})(b\bar{\hat{r}} + \bar{b}\hat{r}) = \frac{1}{4} [ab\bar{\hat{r}}^2 + \bar{a}\bar{b}\hat{r}^2] + \frac{1}{2} (a \cdot b)$$

$$\Rightarrow \quad ab \frac{\bar{r}}{r} + \bar{a}\bar{b} \frac{r}{\bar{r}} = 4(a \cdot \hat{r})(b \cdot \hat{r}) - 2(a \cdot b).$$

Substituting,

$$a_m b_{m'} T^{mm'} = \frac{\pi}{4} \left[-4(a \cdot \hat{r})(b \cdot \hat{r}) + 2(a \cdot b) + 2(a \cdot b) \log \frac{R^2}{r^2} \right]$$

$$T^{mm'} = \frac{\pi}{2} \left(\log \frac{R^2}{r^2} + 1 \right) \delta^{mm'} - \pi \hat{r}^m \hat{r}^{m'}$$

Note where each piece came from: **the logarithm is entirely from the two mixed terms (ii)+(iii)**, and the $\hat{r}\hat{r}$ structure plus the "+1" come from the holomorphic pair (i)+(iv). The +1 is not decoration — it is a genuine finite part and it survives into the physics.

Numerical check ($R = 60$, direct 2D quadrature):

$$r = (0.9, 0): \text{numeric} \begin{pmatrix} 11.6230 & 0 \\ 0 & 14.7645 \end{pmatrix}, \text{exact} \begin{pmatrix} 11.6230 & 0 \\ 0 & 14.7646 \end{pmatrix}.$$

$$r = (-0.6, 1.1): \text{numeric} \begin{pmatrix} 13.008 & 1.314 \\ 1.314 & 11.315 \end{pmatrix}, \text{exact} \begin{pmatrix} 13.005 & 1.321 \\ 1.321 & 11.304 \end{pmatrix} \text{ (max diff 0.012)}.$$

And the trace grows exactly like $\pi \log R^2$: at $R = 20, 60, 180$ the numeric is 19.485, 26.387, 33.290 against 19.485, 26.388, 33.290.

4. Method 2 — $d = 2 - 2\epsilon$ by Feynman parameters

This is the route you will actually use, because $\overline{\text{MS}}$ needs it. Every step below is standard technique worth knowing on its own.

4.1 The Feynman trick

$$\frac{1}{AB} = \int_0^1 \frac{dx}{[xA + (1-x)B]^2}$$

Proof. Write $xA + (1-x)B = B + x(A-B)$. Then $\int_0^1 \frac{dx}{[B+x(A-B)]^2} = \left[\frac{-1}{(A-B)[B+x(A-B)]} \right]_0^1 = \frac{1}{A-B} \left(\frac{1}{B} - \frac{1}{A} \right) = \frac{1}{AB}$ ✓ (verified symbolically, difference 0).

The point is that it converts two different denominators into one, at the price of one extra integration — and the single denominator can then be made spherically symmetric.

4.2 Complete the square

With $A = u^2$ and $B = (u+r)^2$:

$$xu^2 + (1-x)(u+r)^2 = u^2 + (1-x)(2u \cdot r + r^2) = [u + (1-x)r]^2 - (1-x)^2 r^2 + (1-x)r^2$$

$$= \ell^2 + \Delta, \quad \boxed{\ell \equiv u + (1-x)r, \quad \Delta \equiv x(1-x)r^2}$$

since $(1-x) - (1-x)^2 = (1-x)x$. Now the denominator depends only on ℓ^2 .

4.3 Shift the numerator and drop odd terms

From $\ell = u + (1-x)r$: $u = \ell - (1-x)r$ and $u+r = \ell + xr$. So

$$u^m (u+r)^{m'} = \ell^m \ell^{m'} + x \ell^m r^{m'} - (1-x) r^m \ell^{m'} - x(1-x) r^m r^{m'}.$$

The two middle terms are **odd in ℓ** , while the measure $d^d \ell$ and the denominator $(\ell^2 + \Delta)^2$ are even, so they integrate to zero. For the first term use rotational symmetry:

$$\int d^d \ell \ell^m \ell^{m'} f(\ell^2) = \frac{\delta^{mm'}}{d} \int d^d \ell \ell^2 f(\ell^2)$$

(the left side must be $\propto \delta^{mm'}$ since no vector survives; take the trace to fix the coefficient, using $\delta^{mm} = d$). Hence

$$T^{mm'} = \int_0^1 dx \int d^d \ell \frac{\frac{\delta^{mm'}}{d} \ell^2 - x(1-x) r^m r^{m'}}{[\ell^2 + \Delta]^2}.$$

4.4 The Euclidean master integral

$$I_n(\Delta) \equiv \int d^d \ell \frac{1}{(\ell^2 + \Delta)^n} = \pi^{d/2} \frac{\Gamma(n - \frac{d}{2})}{\Gamma(n)} \Delta^{d/2-n}$$

Proof. Go to polar coordinates, $\int d^d \ell = \Omega_d \int_0^\infty \ell^{d-1} d\ell$ with the solid angle $\Omega_d = \frac{2\pi^{d/2}}{\Gamma(d/2)}$. Substitute $t = \ell^2/\Delta$, so $\ell^{d-1} d\ell = \frac{1}{2} \Delta^{d/2} t^{d/2-1} dt$:

$$I_n = \frac{\Omega_d}{2} \Delta^{d/2-n} \int_0^\infty \frac{t^{d/2-1}}{(1+t)^n} dt = \frac{\Omega_d}{2} \Delta^{d/2-n} B\left(\frac{d}{2}, n - \frac{d}{2}\right) = \frac{\pi^{d/2}}{\Gamma(\frac{d}{2})} \Delta^{d/2-n} \frac{\Gamma(\frac{d}{2}) \Gamma(n - \frac{d}{2})}{\Gamma(n)}$$

which is the stated result. (Here $\int_0^\infty \frac{t^{p-1}}{(1+t)^{p+q}} dt = B(p, q)$ is the standard Beta integral.) **Checked symbolically** at $(d, n) = (1, 2), (3, 2), (3, 3), (5, 4)$: difference exactly 0 each time.

4.5 The ℓ^2 numerator

Do not look this up — derive it by adding and subtracting Δ :

$$\begin{aligned} \int d^d \ell \frac{\ell^2}{(\ell^2 + \Delta)^2} &= \int d^d \ell \frac{(\ell^2 + \Delta) - \Delta}{(\ell^2 + \Delta)^2} = I_1 - \Delta I_2 \\ &= \pi^{d/2} \Delta^{d/2-1} \left[\Gamma(1 - \frac{d}{2}) - \Gamma(2 - \frac{d}{2}) \right] = \pi^{d/2} \Delta^{d/2-1} \Gamma(1 - \frac{d}{2}) \left[1 - (1 - \frac{d}{2}) \right] = \pi^{d/2} \frac{d}{2} \Gamma(1 - \frac{d}{2}) \Delta^{d/2-1} \end{aligned}$$

using $\Gamma(2 - \frac{d}{2}) = (1 - \frac{d}{2}) \Gamma(1 - \frac{d}{2})$. (Checked symbolically: difference 0.)

This is where the pole lives. I_1 is $\int \frac{d^d \ell}{\ell^2 + \Delta}$, which diverges at **large** ℓ for $d \geq 2$ — and large ℓ is large u is large $|z|$, exactly the region §2 flagged. Its $\Gamma(1 - \frac{d}{2}) = \Gamma(\epsilon)$ is the collinear pole.

4.6 Assemble, and notice that both x -integrals are the same

Put $d = 2 - 2\epsilon$, so $1 - \frac{d}{2} = \epsilon$ and $2 - \frac{d}{2} = 1 + \epsilon$, and the $\frac{1}{d}$ from §4.3 cancels the $\frac{d}{2}$ from §4.5 down to $\frac{1}{2}$:

$$T^{mm'} = \pi^{d/2} \int_0^1 dx \left\{ \frac{\delta^{mm'}}{2} \Gamma(\epsilon) \Delta^{-\epsilon} - x(1-x) r^m r^{m'} \Gamma(1 + \epsilon) \Delta^{-1-\epsilon} \right\}$$

Now use $\Delta = x(1-x)r^2$ in both places:

$$\Delta^{-\epsilon} = (r^2)^{-\epsilon} [x(1-x)]^{-\epsilon}, \quad x(1-x) \Delta^{-1-\epsilon} = (r^2)^{-1-\epsilon} [x(1-x)]^{-\epsilon}$$

— the explicit $x(1-x)$ in the second term is exactly what is needed to make the two x -integrands identical. So a single Beta function does both:

$$\int_0^1 dx [x(1-x)]^{-\epsilon} = B(1 - \epsilon, 1 - \epsilon) = \frac{\Gamma(1 - \epsilon)^2}{\Gamma(2 - 2\epsilon)}$$

Checked numerically: at $\epsilon = 0.05, 0.20, 0.45$ the quadrature gives 1.106190208, 1.516964233, 2.745420233 against the Gamma expression to 10^{-14} .

Using $\frac{r^m r^{m'}}{r^2} = \hat{r}^m \hat{r}^{m'}$ and $\Gamma(\epsilon) = \Gamma(1 + \epsilon)/\epsilon$:

$$T^{mm'} = \pi^{1-\epsilon} (r^2)^{-\epsilon} \underbrace{\Gamma(1 + \epsilon) \frac{\Gamma(1 - \epsilon)^2}{\Gamma(2 - 2\epsilon)}}_{\mathcal{N}(\epsilon)} \left[\frac{\delta^{mm'}}{2\epsilon} - \hat{r}^m \hat{r}^{m'} \right]$$

Restoring the $\mu^{2\epsilon}$ that the d -dimensional coupling carries, write this as $C_\epsilon \left[\frac{\delta^{mm'}}{2\epsilon} - \hat{r}^m \hat{r}^{m'} \right]$ with $C_\epsilon = \pi \mathcal{N}(\epsilon) (\pi \mu^2 r^2)^{-\epsilon}$.

4.7 Expand in ϵ

$$\Gamma(1 + \epsilon) = 1 - \gamma_E \epsilon + O(\epsilon^2), \quad \frac{\Gamma(1 - \epsilon)^2}{\Gamma(2 - 2\epsilon)} = 1 + 2\epsilon + O(\epsilon^2)$$

$$\implies \mathcal{N}(\epsilon) = 1 + (2 - \gamma_E) \epsilon + O(\epsilon^2), \quad (\pi \mu^2 r^2)^{-\epsilon} = 1 - \epsilon \log(\pi \mu^2 r^2) + O(\epsilon^2)$$

$$\boxed{\frac{C_\epsilon}{\epsilon} = \pi \left[\frac{1}{\epsilon} + 2 - \gamma_E - \log(\pi \mu^2 r^2) \right] + O(\epsilon), \quad C_\epsilon = \pi + O(\epsilon)}$$

$$T^{mm'} = \frac{\pi \delta^{mm'}}{2} \left[\frac{1}{\epsilon} + 2 - \gamma_E - \log(\pi \mu^2 r^2) \right] - \pi \hat{r}^m \hat{r}^{m'} + O(\epsilon)$$

A trap in the expansion. $\frac{\Gamma(1-\epsilon)^2}{\Gamma(2-2\epsilon)}$ is *not* equal to $\frac{1}{1-2\epsilon}$; they agree through $O(\epsilon)$ but differ at $O(\epsilon^2)$ by $-\frac{\pi^2}{6}$. Here only $O(\epsilon)$ is needed, but if you ever push to two loops, do not assume the closed form.

5. The two methods agree — and that is a real check

Compare §3.6 with §4.7:

$$\log \frac{R^2}{r^2} + 1 \longleftrightarrow \frac{1}{\epsilon} + 2 - \gamma_E - \log(\pi \mu^2 r^2) \implies \log R^2 = \frac{1}{\epsilon} + 1 - \gamma_E - \log(\pi \mu^2).$$

That is just a dictionary between two regulators and carries no physics. **What does carry physics is the $\hat{r}^m \hat{r}^{m'}$ coefficient: it is $-\pi$ in both.** It is finite, so it cannot depend on how you regulated; getting the same number from a contour argument in $d = 2$ and from Gamma functions in $d = 2 - 2\epsilon$ is a genuine, independent check that both calculations are right. The same goes for the relative "+1" in the $\delta^{mm'}$ piece.

6. $\overline{\text{MS}}$

The $\overline{\text{MS}}$ scheme subtracts not just the pole but the universal $-\gamma_E + \log 4\pi$ that always rides along with it:

$$\frac{1}{\epsilon_{\overline{\text{MS}}}} \equiv \frac{1}{\epsilon} - \gamma_E + \log 4\pi.$$

Rewrite the bracket, adding and subtracting $\log 4\pi$:

$$\frac{1}{\epsilon} + 2 - \gamma_E - \log(\pi \mu^2 r^2) = \underbrace{\left[\frac{1}{\epsilon} - \gamma_E + \log 4\pi \right]}_{1/\epsilon_{\overline{\text{MS}}}} + 2 - \underbrace{\left[\log 4\pi + \log(\pi \mu^2 r^2) \right]}_{\log(4\pi^2 \mu^2 r^2)}$$

$$\boxed{\mathcal{L} \equiv \frac{1}{\epsilon_{\overline{\text{MS}}}} - \log(4\pi^2 \mu^2 r^2), \quad T^{mm'} = \frac{\pi}{2} (\mathcal{L} + 2) \delta^{mm'} - \pi \hat{r}^m \hat{r}^{m'}}$$

Renormalisation then removes the $\frac{1}{\epsilon_{\overline{\text{MS}}}}$ — here into the gluon fragmentation function $D_{g/g}(\zeta, \mu)$ — leaving the physical $\mathcal{L} \rightarrow -\log(4\pi^2 \mu^2 r^2)$, whose μ -dependence is then cancelled by the DGLAP evolution of D .

7. The one trap that will bite you: never set $d_\perp = 2$ too early

$T^{mm'}$ carries a $\frac{1}{\epsilon}$. If you contract it with anything that itself depends on ϵ , the $O(\epsilon)$ part of that thing hits the pole and leaves a **finite** remainder. In your row this happens in exactly one place, and it changes the answer.

7.1 Contracting with $\delta_{mm'}$ — the "+2" gets eaten

The trace must be taken in $d_\perp = 2 - 2\epsilon$ dimensions, not 2:

$$\text{tr } T = \delta_{mm'} T^{mm'} = \frac{\pi}{2} (\mathcal{L} + 2) d_\perp - \pi = \frac{\pi}{2} (\mathcal{L} + 2)(2 - 2\epsilon) - \pi$$

Now $\mathcal{L}$ contains $\frac{1}{\epsilon}$, so the -2ϵ multiplying it leaves behind a finite $-\frac{\pi}{2} \cdot 2\epsilon \cdot \frac{1}{\epsilon} = -\pi$:

$$\text{tr } T = \pi(\mathcal{L} + 2) - \pi - \pi = \boxed{\pi \mathcal{L}}$$

The "+2" has been eaten. Had you set $d_\perp = 2$ first you would have got $\pi(\mathcal{L} + 2)$ — wrong by 2π in the finite part.

7.2 Contracting with $X^m X'^{m'}$ — the "+2" survives

Here $X^k = \frac{(x-y)^k}{(x-y)^2}$ and $X'^{k'} = \frac{(x'-y')^{k'}}{(x'-y')^2}$ carry no ϵ at all, so nothing is eaten:

$$X_m X'_m T^{mm'} = \frac{\pi}{2}(\mathcal{L} + 2) \mathcal{X} - \pi \mathcal{R}, \quad \mathcal{X} \equiv X \cdot X', \quad \mathcal{R} \equiv (X \cdot \hat{r})(X' \cdot \hat{r}).$$

Rule of thumb. Keep $d_\perp$ symbolic, contract, expand in ϵ , and only then set $d_\perp = 2$. Setting it early is safe only for structures with no free ϵ -dependence multiplying the pole.

8. Back into the row

Your post- w bracket is $b_1 \delta_{km} \delta_{k'm'} + b_2 \delta_{mm'} \delta_{kk'} + b_3 \delta_{mk'} \delta_{km'}$ with $b_1 = \frac{d_\perp p^+}{S^2} - \frac{2}{k^+}$, $b_2 = \frac{p^+}{k^{+2}} + \frac{1}{p^+}$, $b_3 = \frac{2}{k^+}$. Contracting with $T^{mm'} X^k X'^{k'}$:

- b_1 and b_3 both hit $X_m X'_m T^{mm'}$ (the b_3 one because T is symmetric), so they add: $b_1 + b_3 = \frac{d_\perp p^+}{S^2}$. The $\frac{2}{k^+}$ **cancels** — that cancellation is the momentum sum rule $\zeta + \bar{\zeta} = 1$.
- b_2 hits $(\text{tr } T) \mathcal{X} = \pi \mathcal{L} \mathcal{X}$ — §7.1.

$$\frac{d_\perp p^+}{S^2} \left[\frac{\pi}{2}(\mathcal{L} + 2) \mathcal{X} - \pi \mathcal{R} \right] + \left(\frac{p^+}{k^{+2}} + \frac{1}{p^+} \right) \pi \mathcal{L} \mathcal{X}$$

and the $d_\perp = 2 - 2\epsilon$ in the first term eats one more unit from its own $\mathcal{L} + 2$, turning it into $\mathcal{L} + 1$. Writing $\zeta = \frac{k^+}{S}$, $\bar{\zeta} = \frac{p^+}{S}$, so that $\frac{p^+}{S^2} = \frac{\zeta \bar{\zeta}}{k^+}$, $\frac{p^+}{k^{+2}} = \frac{\bar{\zeta}}{\zeta k^+}$, $\frac{1}{p^+} = \frac{\zeta}{\bar{\zeta} k^+}$, everything collapses:

$$\frac{\pi}{k^+} \left\{ \underbrace{\left[\zeta \bar{\zeta} + \frac{\bar{\zeta}}{\zeta} + \frac{\zeta}{\bar{\zeta}} \right]}_{C_{UV}(\zeta) = P_{gg}(\zeta)/2N_c} \mathcal{L} \mathcal{X} + \zeta \bar{\zeta} [\mathcal{X} - 2\mathcal{R}] \right\}$$

Verified symbolically against the direct ϵ -expansion: difference exactly 0. That C_{UV} multiplies the logarithm — and only the logarithm — is the whole DGLAP content of the row.

9. Checklist for next time

1. Shift so the integral depends on one vector only. Note its symmetry and write down the two tensor structures it can have.
2. **Power count before integrating.** Find which region diverges. That fixes the regulator and the sign of ϵ .
3. If $d = 2$ and you want intuition: complex coordinates, $\frac{a \cdot u}{u^2} = \frac{1}{2} \left(\frac{a}{u} + \frac{\bar{a}}{\bar{u}} \right)$, and the two masters of §3.3–3.4.
4. If you need $\overline{\text{MS}}$: Feynman parameter, complete the square, shift, drop odd terms, $\ell^m \ell^{m'} \rightarrow \frac{\delta^{mm'}}{d} \ell^2$, then I_n and a Beta function.
5. Cross-check the two against each other on the **finite** structure — it must be regulator-independent.
6. Contract with $d_\perp$ symbolic. Expand. Then set $d_\perp = 2$.

Every step above is checked in `src/row23_zint.py`; output in `RESULTS_row23.txt`.